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Calibrating Decision Robustness via Inverse Conformal Risk Control

Tracing the Miscalibration-Regret Pareto Frontier Instead of Fixing a Coverage Target in Advance

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Robust optimization protects decisions by hedging against a conformal uncertainty set—but the robustness level behind that set is usually fixed ad hoc, either exposing decisions or wasting performance on needless conservatism. **Inverse Conformal Risk Control (ICRC)** flips the usual recipe: instead of fixing coverage and later measuring regret, it delivers simultaneous, distribution-free, finite-sample guarantees on both miscalibration *and* regret across an entire family of policies, letting decision-makers pick their robustness level directly off a certified Pareto frontier.

AT A GLANCE

Problem: The coverage level behind a robust conformal decision policy is normally fixed a priori, with no statistical way to judge whether it was the right choice.

Why it matters: As conformal sets increasingly feed robust optimization pipelines in finance, operations, and safety-critical planning, an unexamined robustness knob leaves performance on the table invisibly.

Method: Simultaneous, distribution-free finite-sample bounds on both miscalibration and regret across a family of robustness levels, valid uniformly.

Result: A certified miscalibration-regret Pareto frontier that decision-makers can select from *after* seeing it, with the guarantee still holding.

Evidence: Robust predict-then-optimize benchmarks comparing ICRC’s certified frontier against fixed- α conformal robust optimization.

The Big Picture

In robust predict-then-optimize, a conformal uncertainty set $\mathcal{U}_\alpha(x)$ is built at coverage level α , and a decision minimizes the worst-case cost over that set. The choice of α determines both the protection the decision receives and the optimization performance (regret) sacrificed for that protection—but conventional practice fixes α once, in advance, with no principled way to know whether that choice was too cautious or not cautious enough for the task’s

real cost structure. This paper makes the entire tradeoff between protection and performance a statistically certified, inspectable object rather than a single unexamined hyperparameter.

Why Existing Approaches Fall Short

Fixing a single coverage target α and only afterward measuring the regret it induces means the decision-maker learns whether their choice was good only after committing to it. Re-running the pipeline for a different α offers no statistical guarantee that the new choice is valid, since the original calibration was only certified for the original α . This traps practitioners into a single ad hoc choice with no honest way to compare alternatives.

Core Mechanism

Rather than treating α as a knob set once, ICRC treats the entire curve relating miscalibration to regret as the object of interest. It constructs a family of policies indexed by robustness level and provides distribution-free, finite-sample bounds on both the true miscalibration rate and the true regret at every level in that family *simultaneously*, so a decision-maker can inspect the empirical Pareto frontier after the fact and select whichever point matches their actual cost-risk preference—with the guarantee still holding for that a posteriori choice.

Technical Formulation

For a family of robust policies $\{\pi_\alpha\}$ built from uncertainty sets $\mathcal{U}_\alpha(x)$, define

$$\text{miscalibration}(\alpha) = P(y \notin \mathcal{U}_\alpha(x)),$$

$$\text{regret}(\alpha) = \mathbb{E}[\text{cost}(\pi_\alpha(x), y)] - \min_a \mathbb{E}[\text{cost}(a, y)].$$

These move in opposite directions as α varies, forming a Pareto frontier. ICRC constructs simultaneous confidence bands for both functions, valid uniformly over a grid of α values with probability at least $1 - \delta$, using conformal calibration on a held-out set combined with uniform concentration arguments (in the style of DKW-type bounds) to control the multiple-testing-like inflation from evaluating many α values on the same data.

Learning or Inference Procedure

1. Build a grid of candidate robustness levels α and their corresponding uncertainty sets $\mathcal{U}_\alpha(x)$.
2. On held-out calibration data, estimate miscalibration and regret for every α in the grid.
3. Apply uniform concentration bounds to certify simultaneous validity of both estimates across the whole grid.

4. Present the certified empirical Pareto frontier; the decision-maker selects $\hat{\alpha}$ off the frontier according to their cost-risk preference.

What the Guarantee Says

With probability at least $1 - \delta$, the true miscoverage and true regret curves are simultaneously bracketed by the certified bands at every α on the grid—so whichever $\hat{\alpha}$ the decision-maker picks *after* inspecting the frontier still enjoys valid finite-sample guarantees on both quantities, unlike a guarantee calibrated for one α chosen in advance.

Experimental Findings

On robust predict-then-optimize benchmarks, ICRC’s certified Pareto frontier lets practitioners identify robustness levels that achieve materially lower regret than conventional fixed- α conformal robust optimization at matched miscoverage, and vice versa—revealing that ad hoc α choices in prior pipelines were leaving performance on the table without practitioners ever being able to see it.

Ablations and Interpretation

Grid resolution: Finer grids over α tighten the achievable frontier at a modest cost in the width of the simultaneous confidence bands, reflecting the usual multiple-testing tradeoff.

Comparison to fixed- α baselines: Any single fixed- α policy corresponds to one point on ICRC’s frontier; the benefit is purely in being able to see and choose among the alternatives with statistical validity.

Practical takeaway: ICRC does not require the analyst to predict the right robustness level in advance—it makes the tradeoff observable and lets the choice come last.

Reference

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